

UNDERSTANDING GRAMMAR REPRESENTATION IN THE HUMAN BRAIN VIA INFORMATION-RESTRICTED LLMS

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How does the brain distinguish nouns and verbs?

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Fronto-Temporal Dichotomy

Damasio and Tranel (1993)

Patients with **brain lesions** doing **object** and **action** naming tasks

→ **Temporal** regions involved in the retrieval of **nouns**

→ **Frontal** regions involved in the retrieval of **verbs**

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What is driving the distinction?

How does the brain distinguish nouns and verbs? **It's complicated...**

Emergentist View

Bird et al. (2000), Vigliocco et al. (2011)

Noun / Verb distinction is not **lexical** but **semantic**

→ Nouns are **more imageable** and related to **objects**

→ Verbs are **less imageable** and related to **actions**

How does the brain distinguish nouns and verbs? **It's complicated...**

Combinatorial View

Shapiro and Caramazza (2003), Tyler et al. (2003)

Noun / Verb distinction driven by **computational** processes

→ Pseudowords: “*he wugged*” and “*the wugs*” represented in distinct brain areas

→ Verbs are more **computationally demanding**

How does the brain distinguish nouns and verbs? **It's complicated...**

Distributed Account

Crepaldi et al. (2013)

Theory-blind meta-analysis based on 36 neuroimaging studies

→ 32 reliable **clusters**:

3 for **nouns** (R-ITG, L-AG, L-IPG)

1 for **verbs** (R-MTG)

3 both **nouns** and **verbs**; Task dependent (L-IFG, R-IFG, L-Insula)

→ No support for **Fronto-Temporal** dichotomy or **Embodied Cognition**

→ **Distributed** processing of nouns and verbs

Methods

Picture Naming (Shapiro and Caramazza, 2011)

Semantic Categorization (Tyler et al., 2003)

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Passive processing?

Methods

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Voxel-wise Encoding Models

Stimuli

11 stories from *The Moth Radio Hour*

10-15 minutes per story

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fMRI Data

9 Subjects
Listening and reading Data

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Listening and reading Data

Features

English1000, word count, pauses

Reading: letters, letter count

Listening: phonemes, phoneme count

Stimulus

There're some scientists who say that the universe that we inhabit is not the only universe there is ...

Stimulus

*There're some scientists who say that the universe
that we inhabit is not the only universe there is ...*

↓ POS tagging via spaCy

Attention Mask

There're	some	scientists	who	say	that	the	universe		
that	we	inhabit	is	not	the	only	universe	there	is

Stimulus

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GPT-2

Layer 1

⋮

Layer 8

⋮

Layer 12

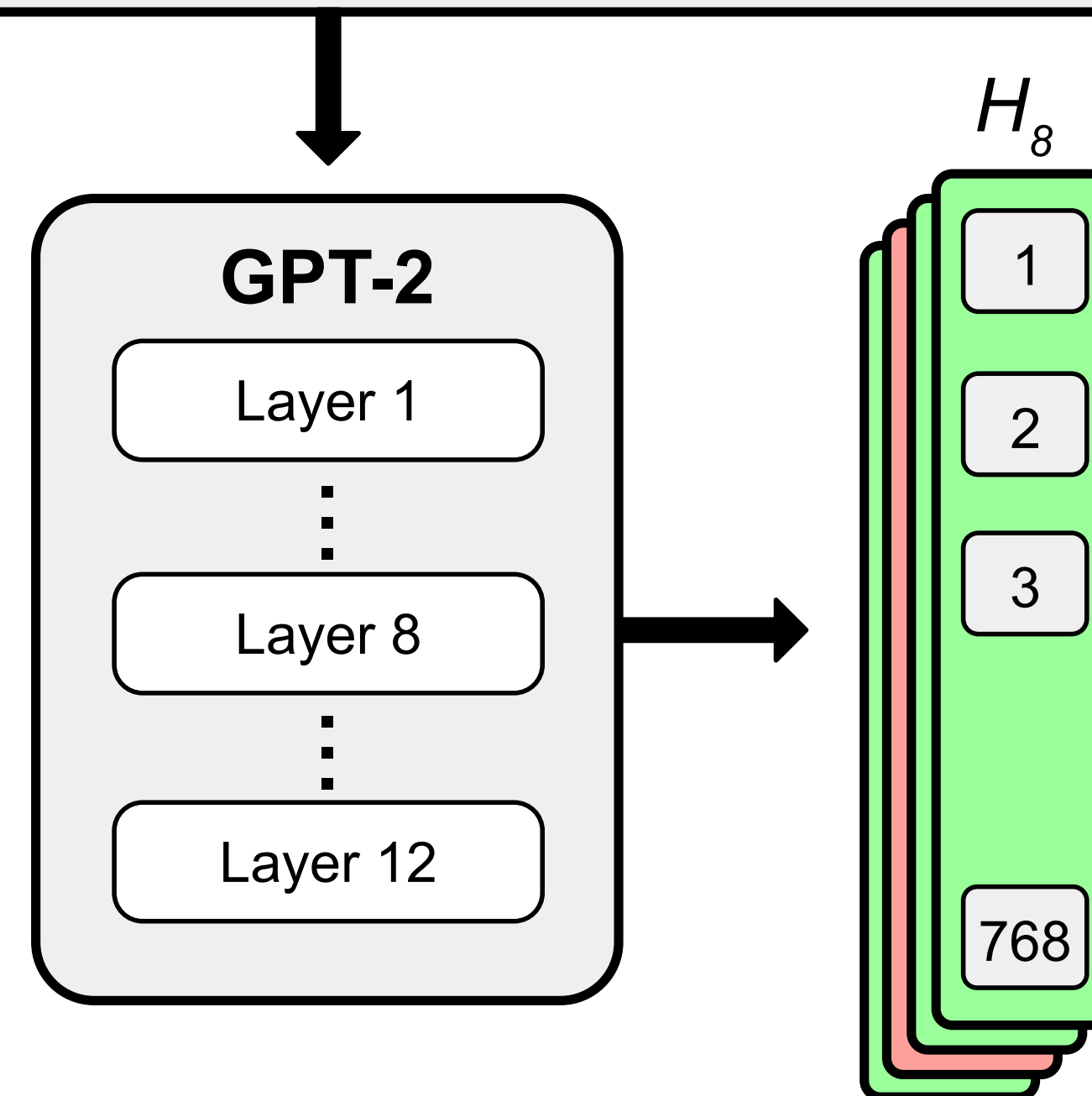
Stimulus

*There're some scientists who say that the universe
that we inhabit is not the only universe there is ...*

↓ POS tagging via spaCy

Attention Mask

*There're some scientists who say that the universe
that we inhabit is not the only universe there is*



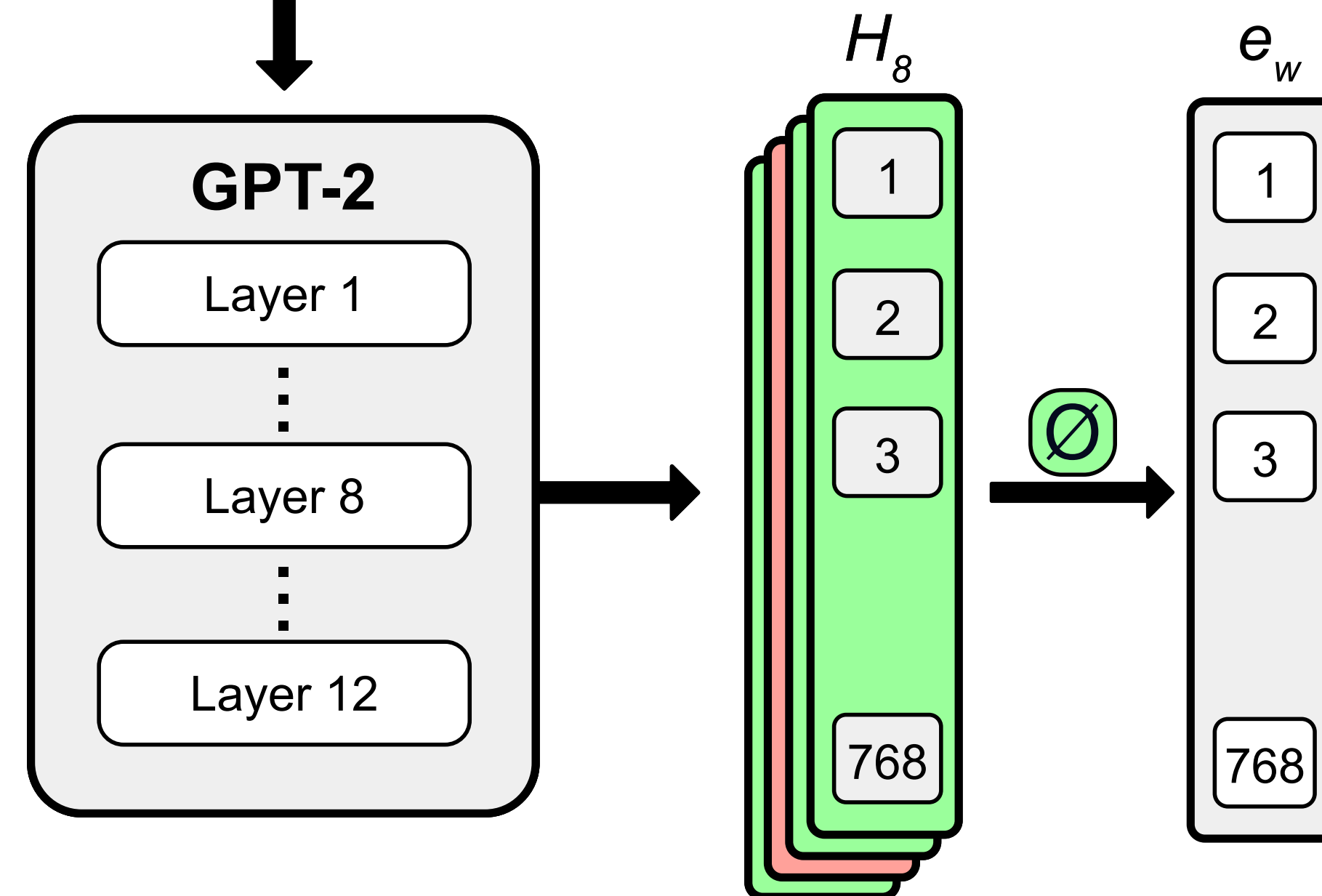
Stimulus

*There're some scientists who say that the universe
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Banded Ridge Regression

to model BOLD as linear combination of masked embeddings and low level features

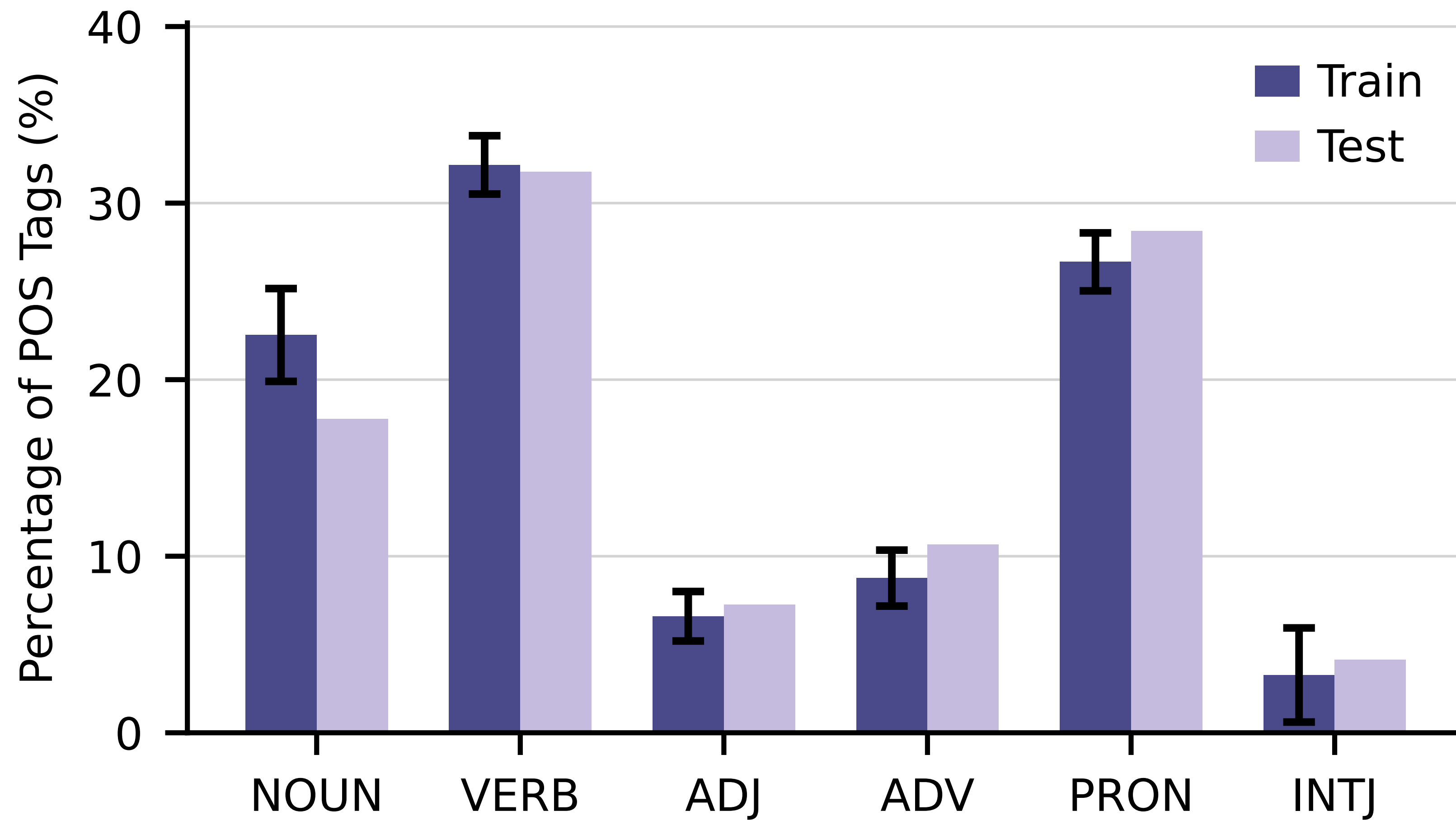
POS used

nouns, verbs, adjectives, adverbs, pronouns, interjections

Prediction accuracy

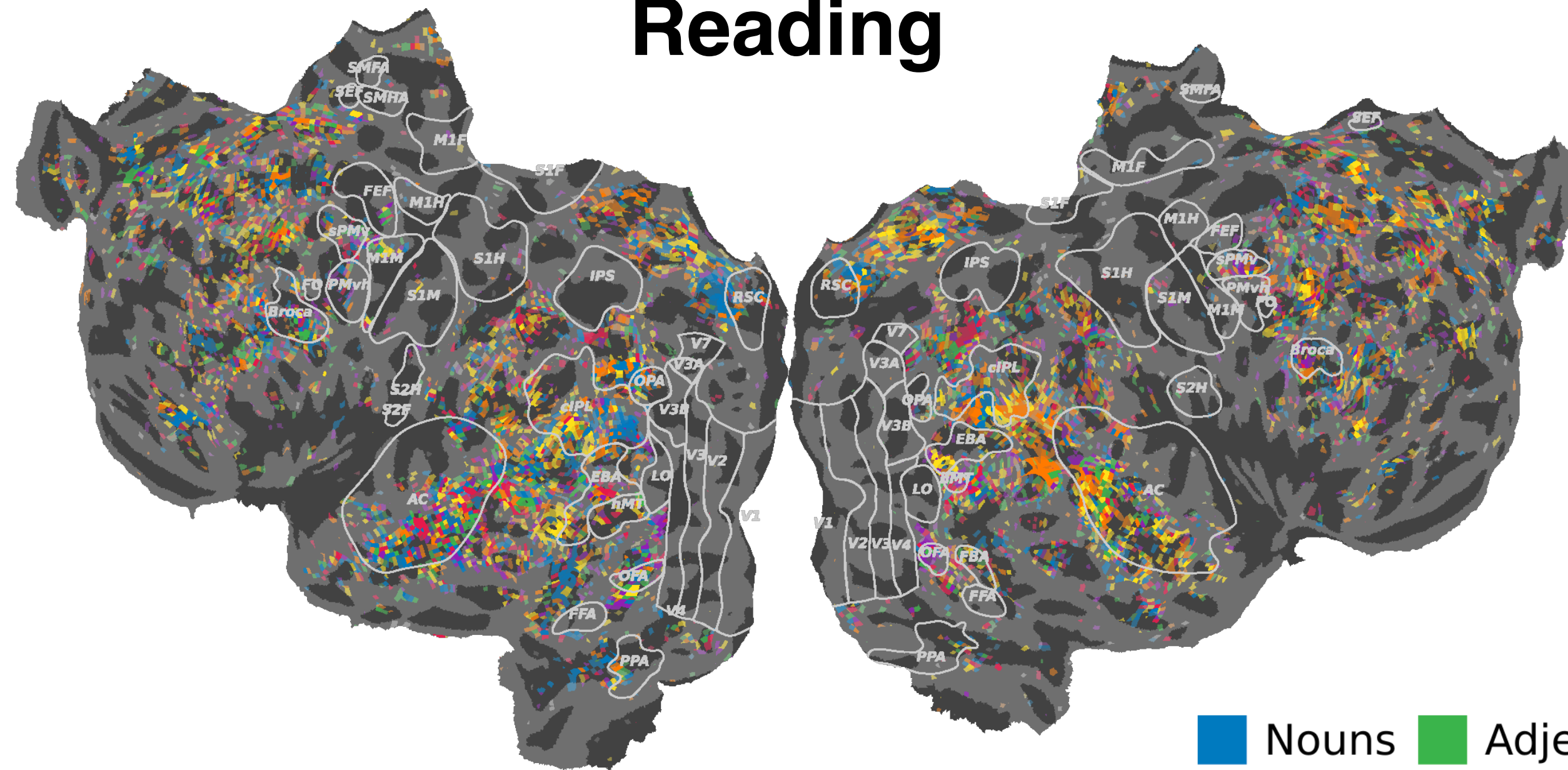
Pearson correlation between predicted and actual train data BOLD response

PARTS OF SPEECH DISTRIBUTION IN DATASET

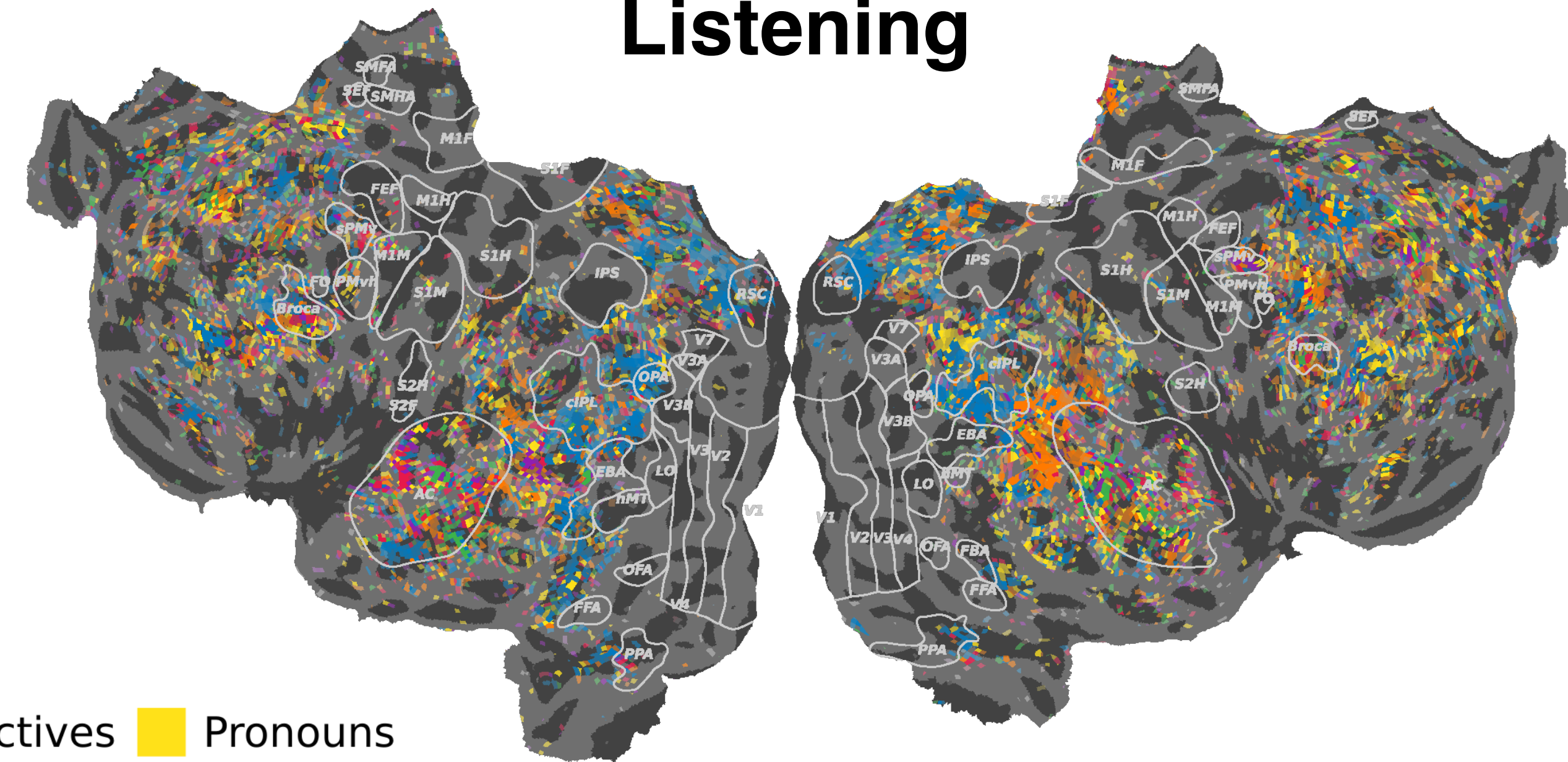


PART-OF-SPEECH SENSITIVITY

Subject 7
Reading



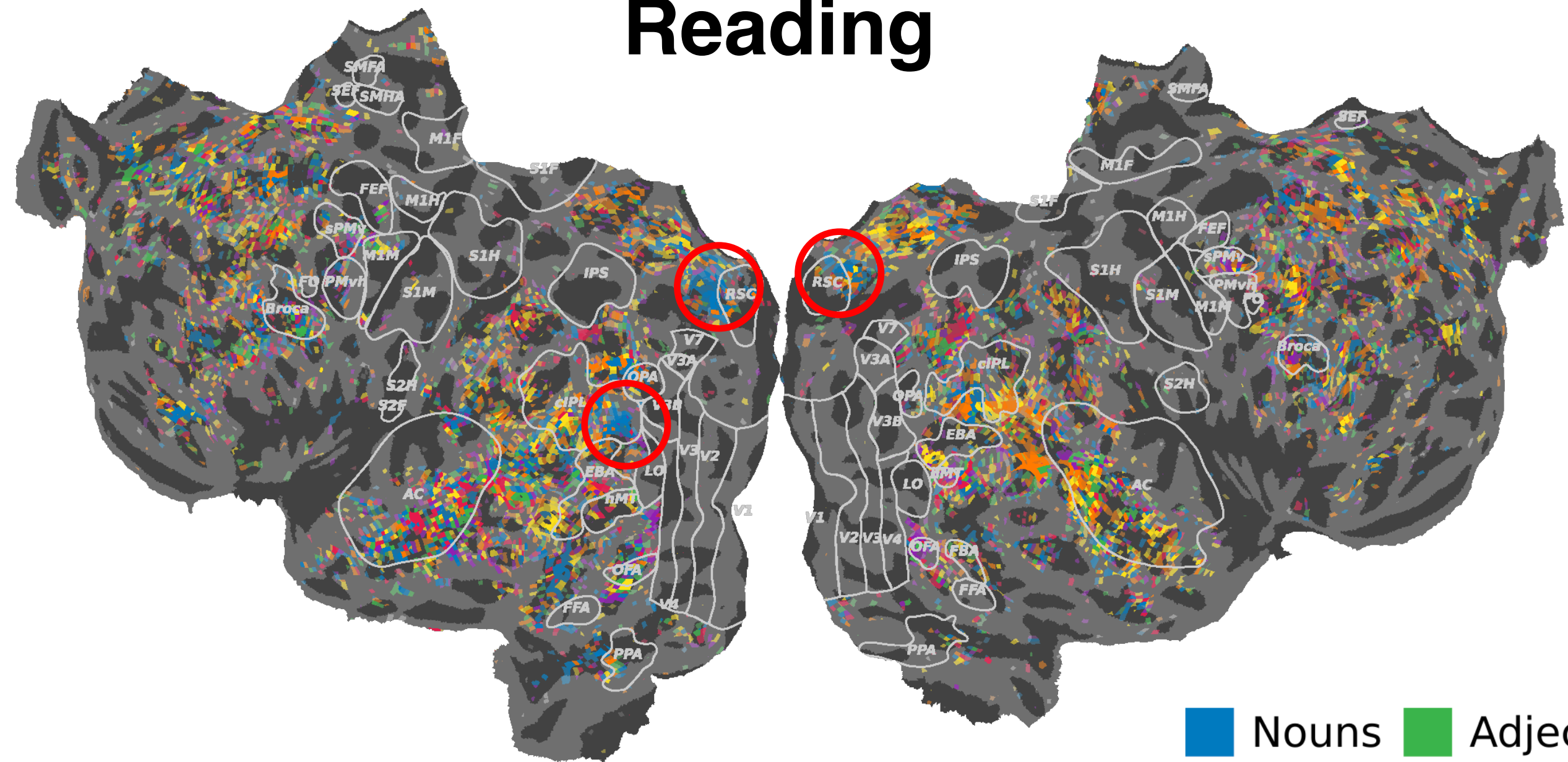
Subject 7
Listening



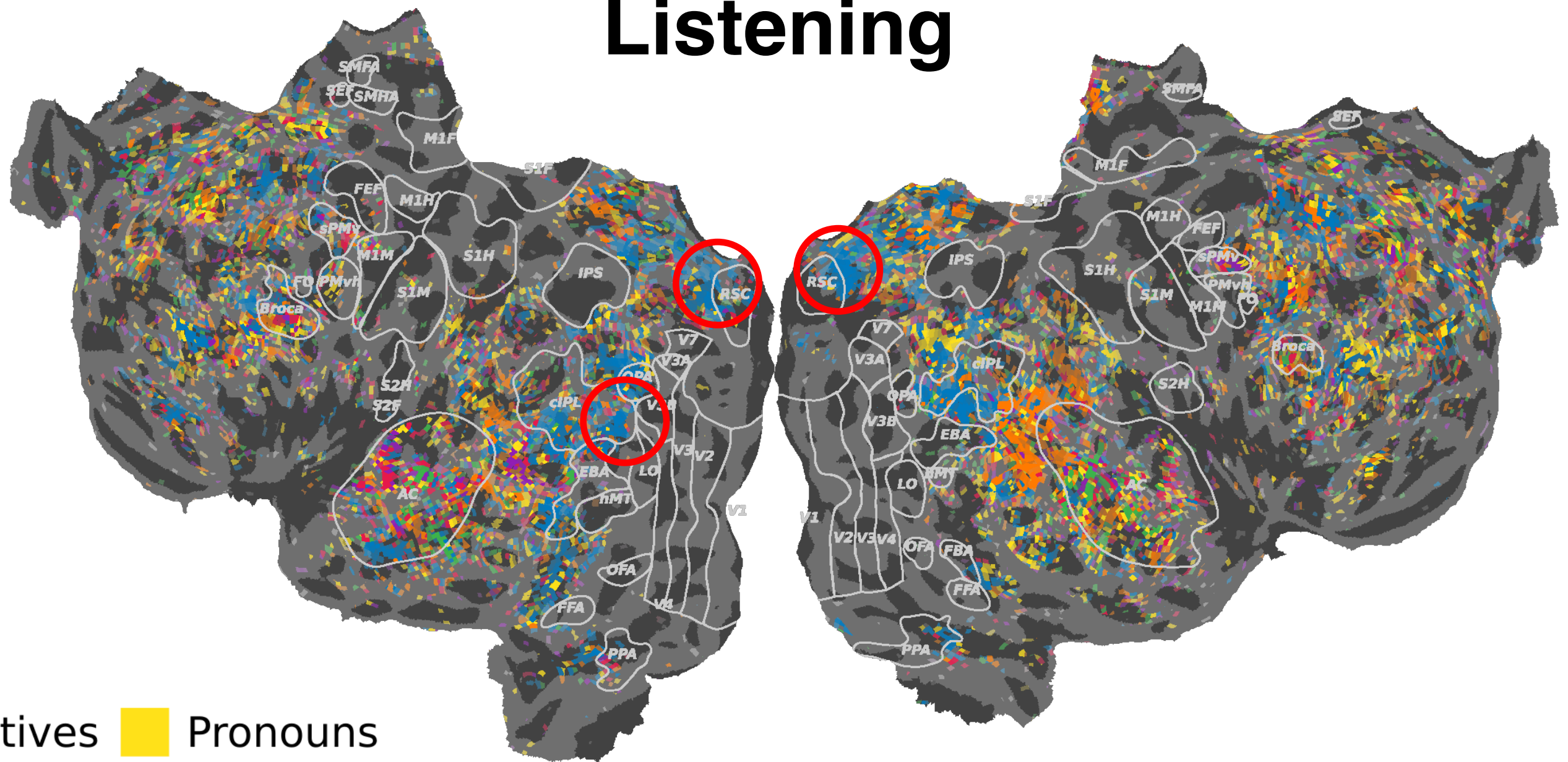
■ Nouns ■ Adjectives ■ Pronouns
■ Verbs ■ Adverbs ■ Interjections

PART-OF-SPEECH SENSITIVITY

Subject 7
Reading



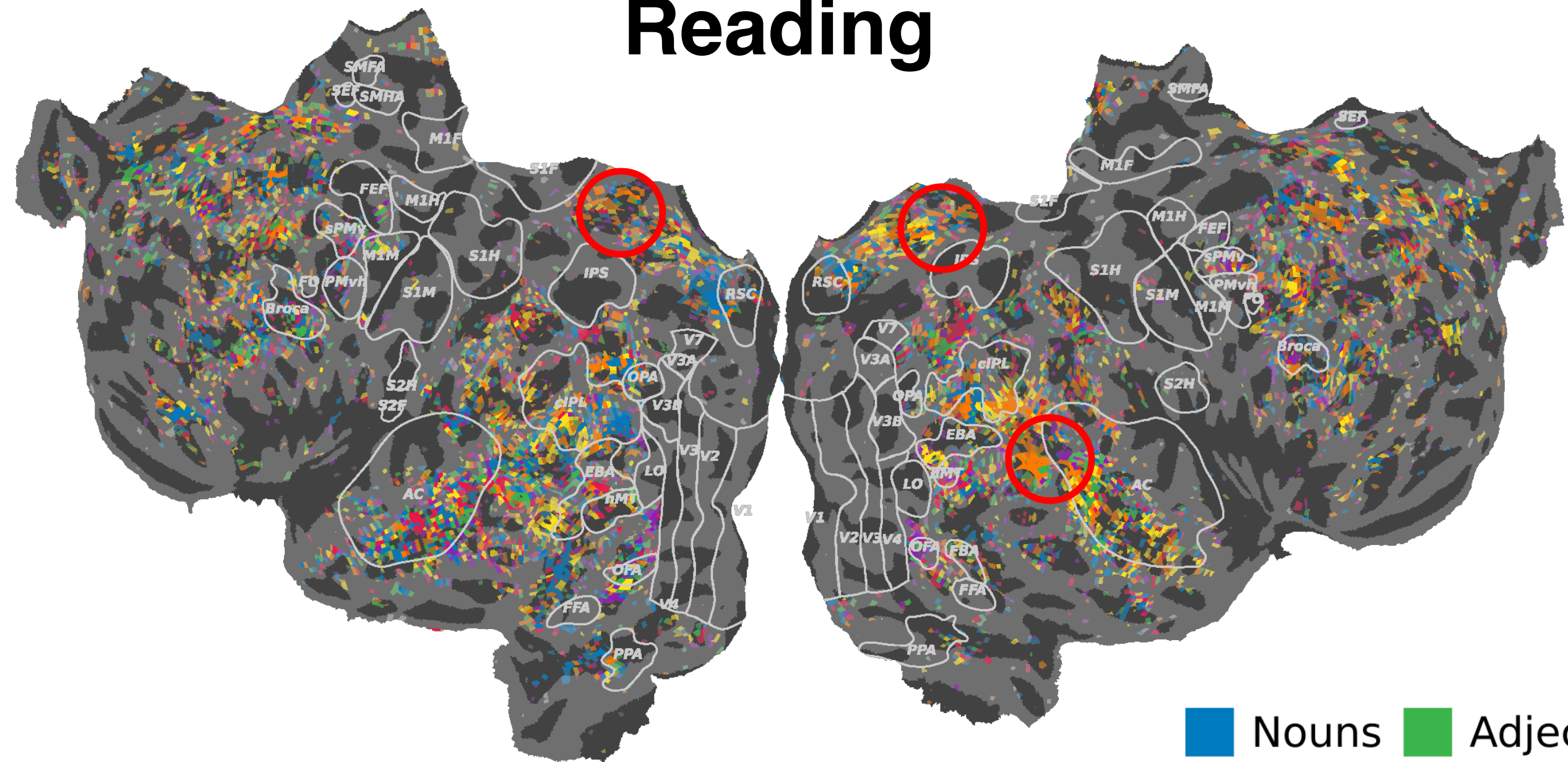
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Listening



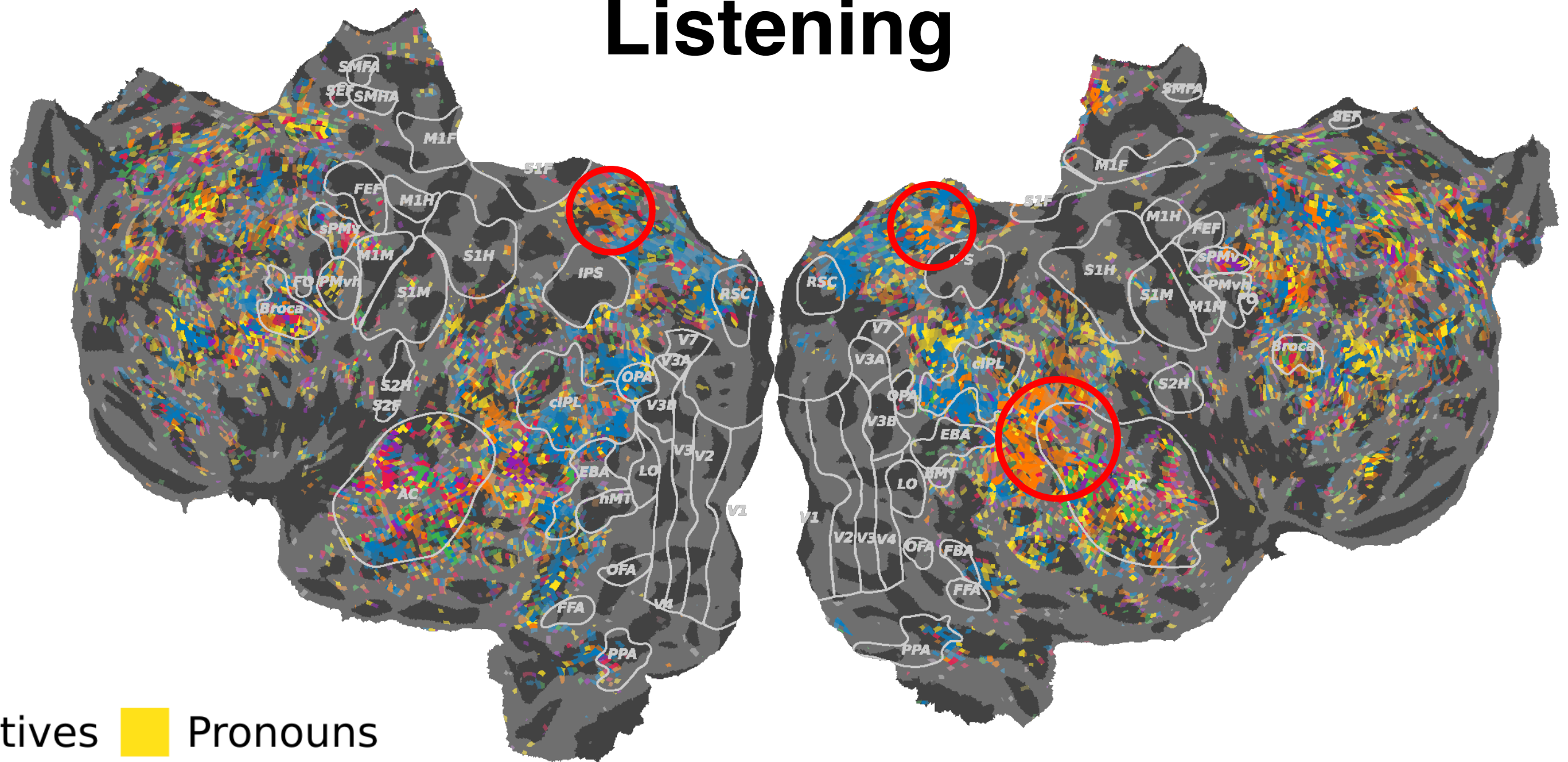
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PART-OF-SPEECH SENSITIVITY

Subject 7
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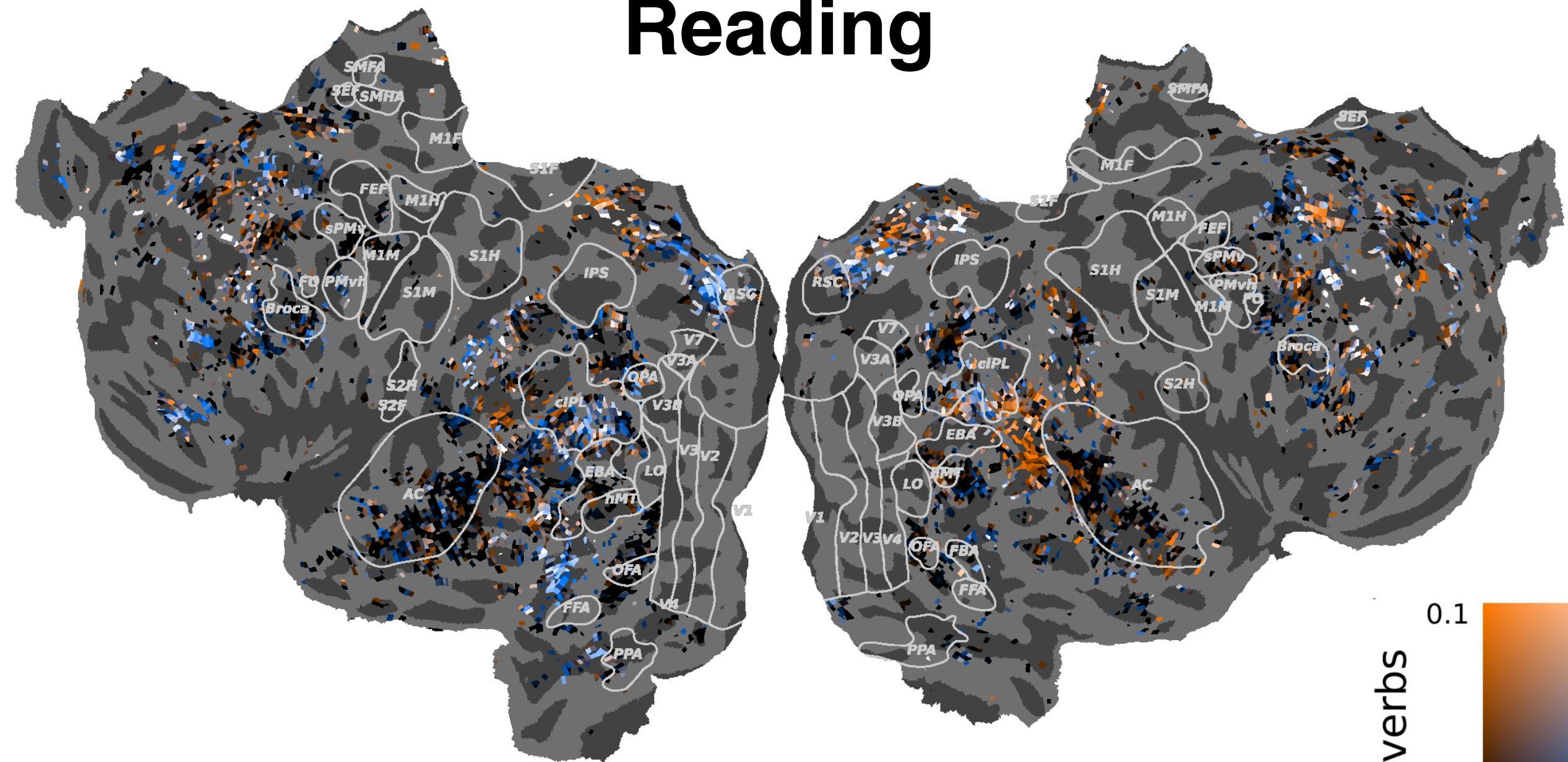
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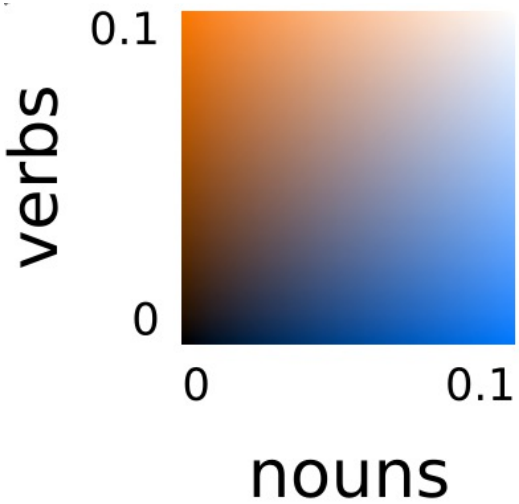
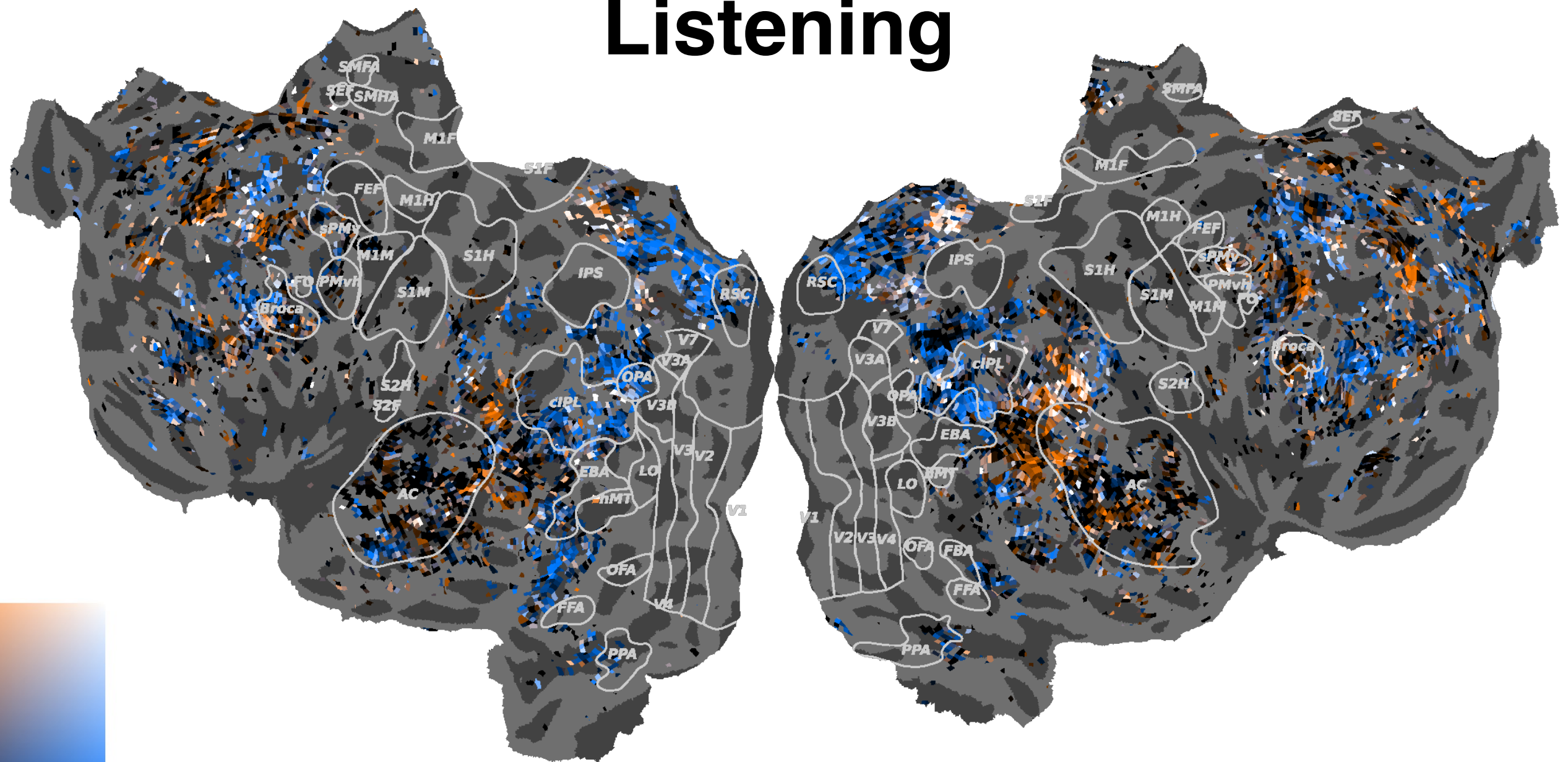
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NOUN & VERB SENSITIVITY

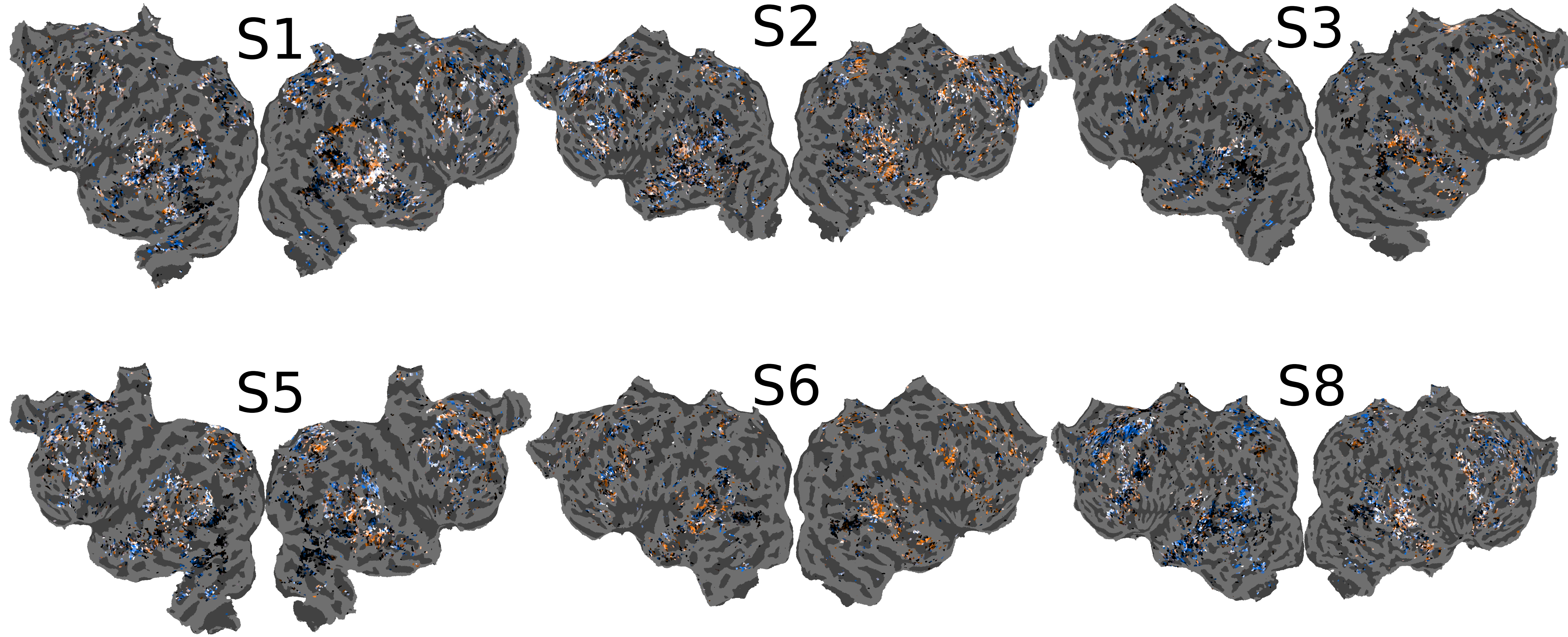
Subject 7
Reading



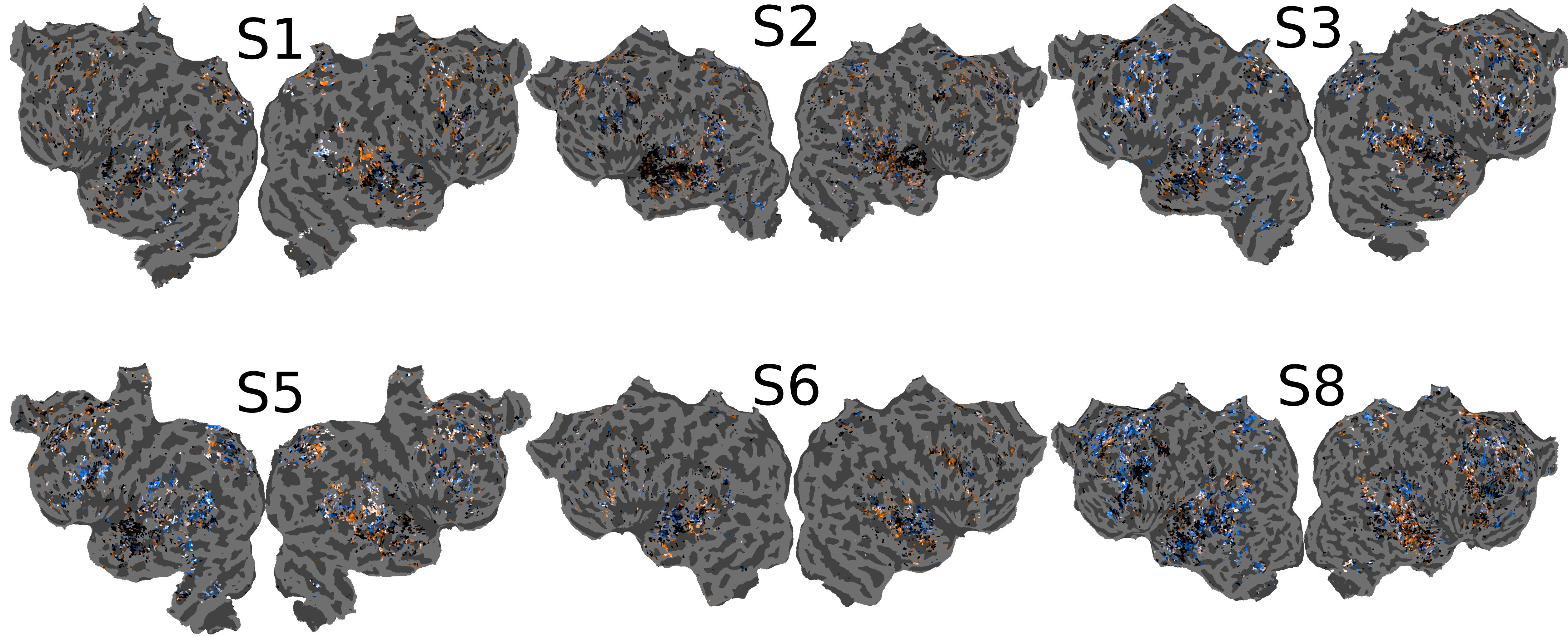
Subject 7
Listening



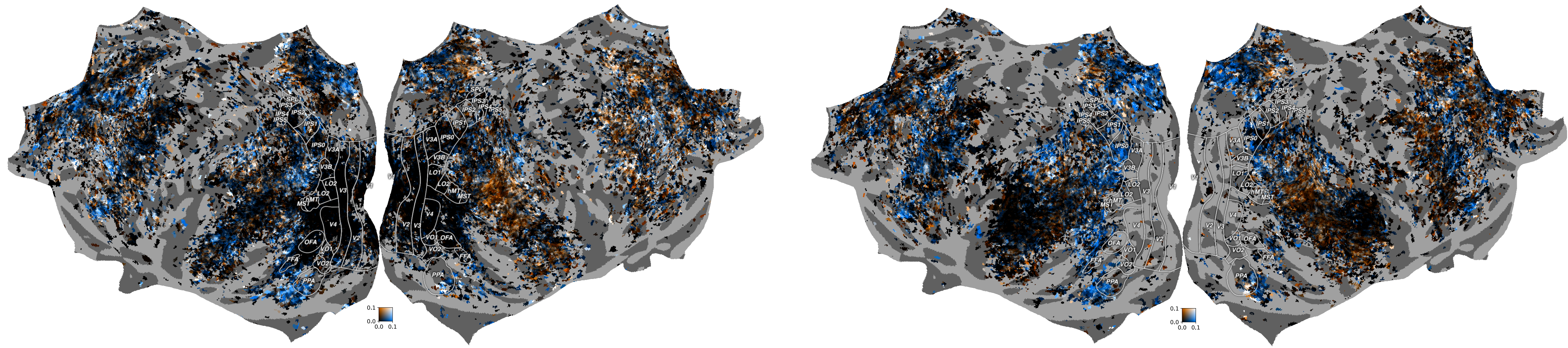
NOUN & VERB SENSITIVITY - READING



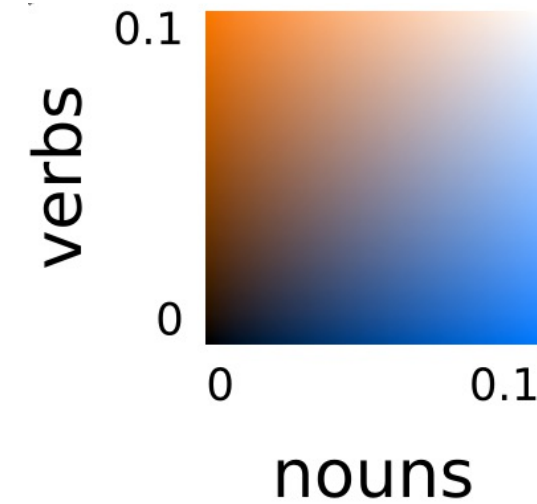
NOUN & VERB SENSITIVITY - LISTENING



NOUN & VERB SENSITIVITY ACROSS SUBJECTS

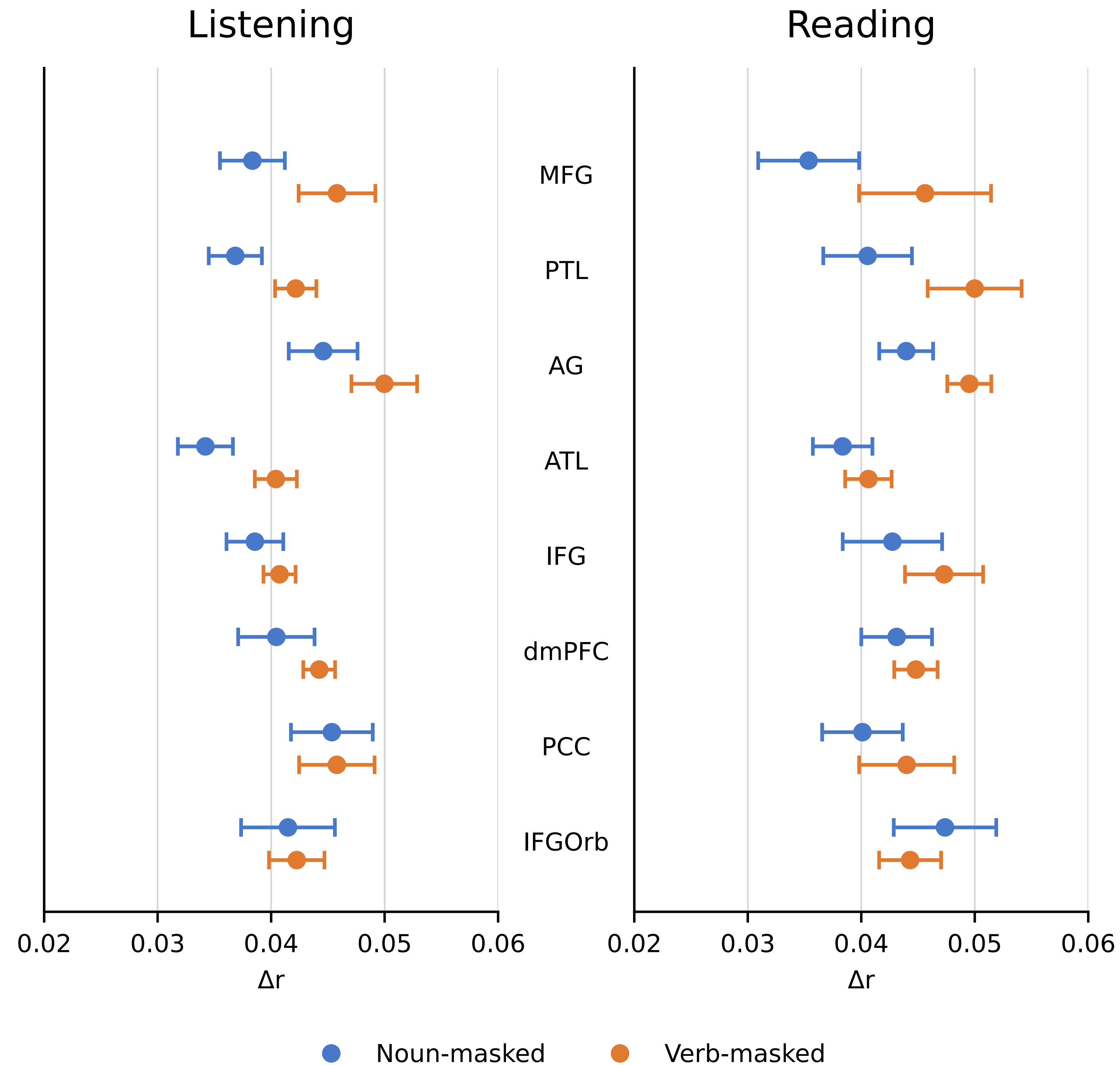


**Across Subjects
Reading Modality**



**Across Subjects
Listening Modality**

ROI ANALYSIS



Language ROIs

Fedorenko et al. (2010)

Mean Drop

0.03 – 0.06

Verb > Noun

MFG, PTL, AG, ATL

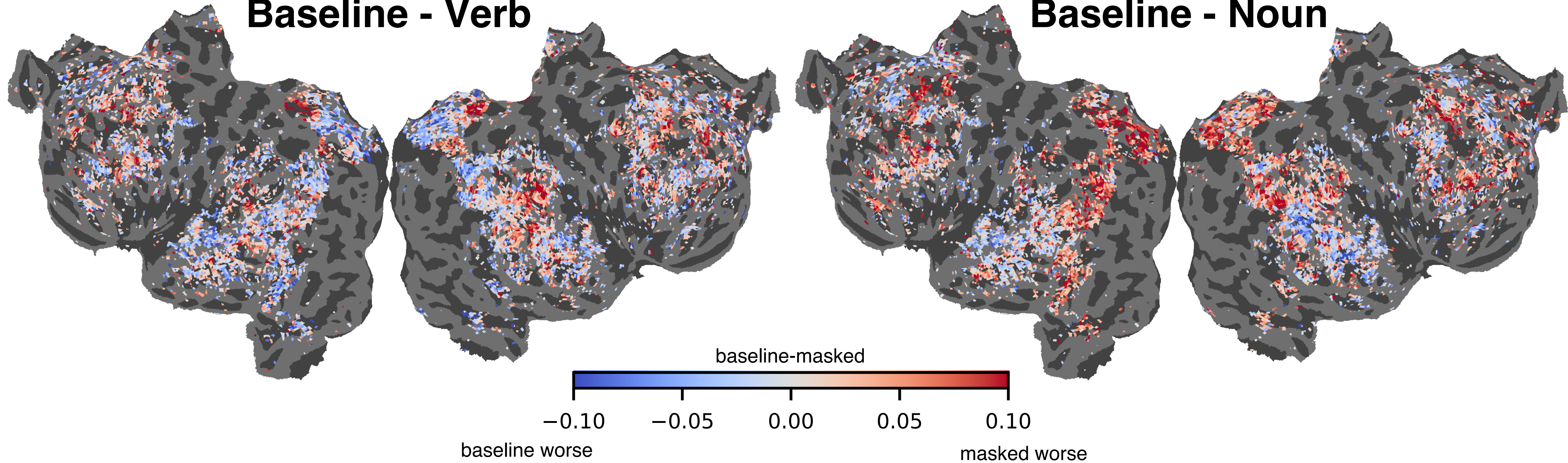
Reading vs Listening

Similar across ROIs

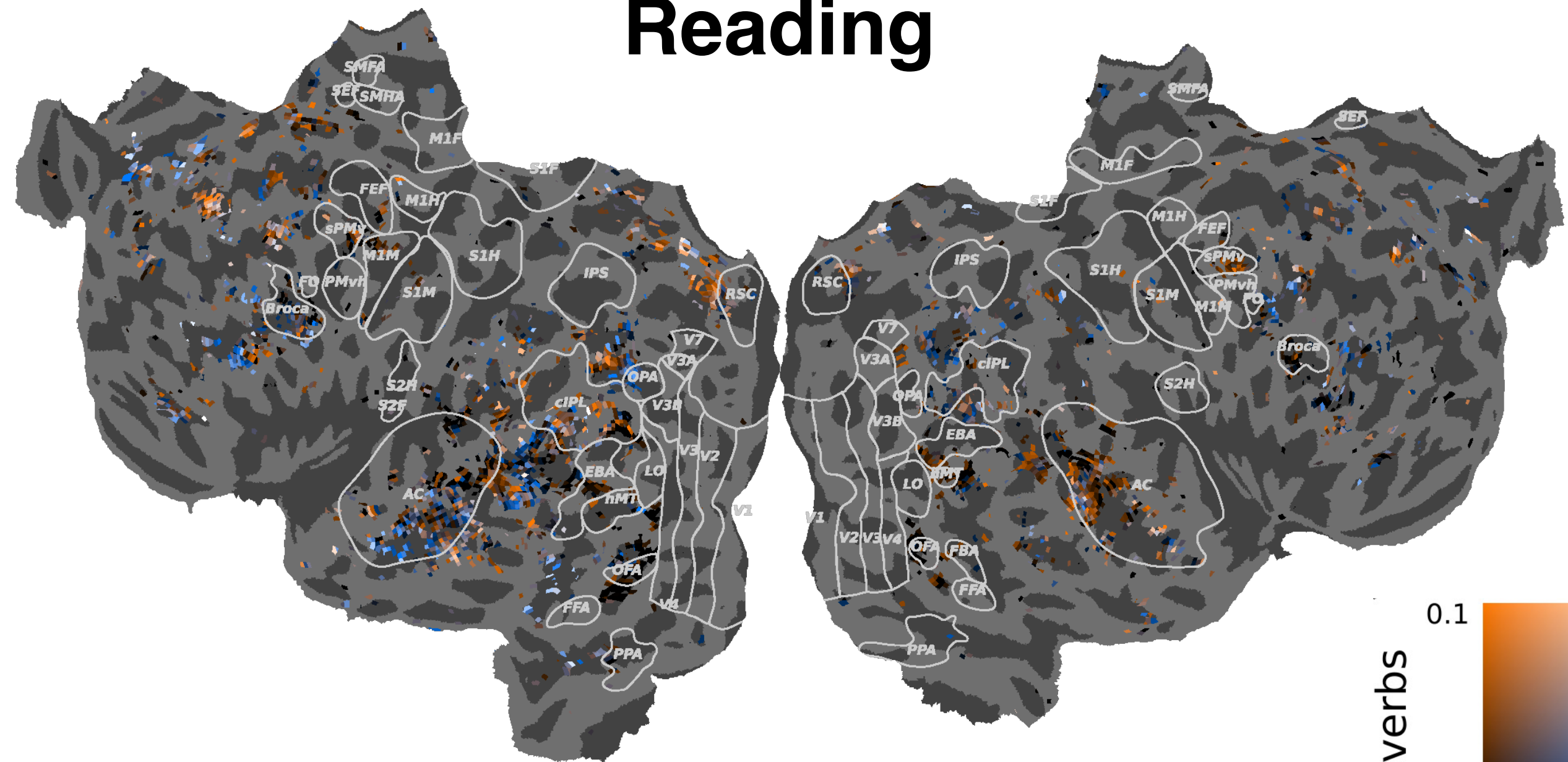
BASELINE CONTRAST

Subject 7
Baseline - Verb

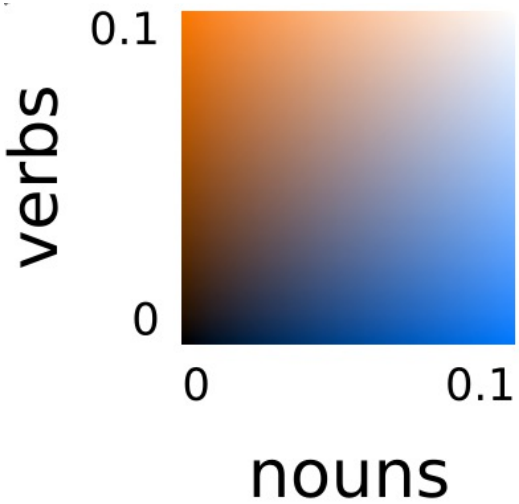
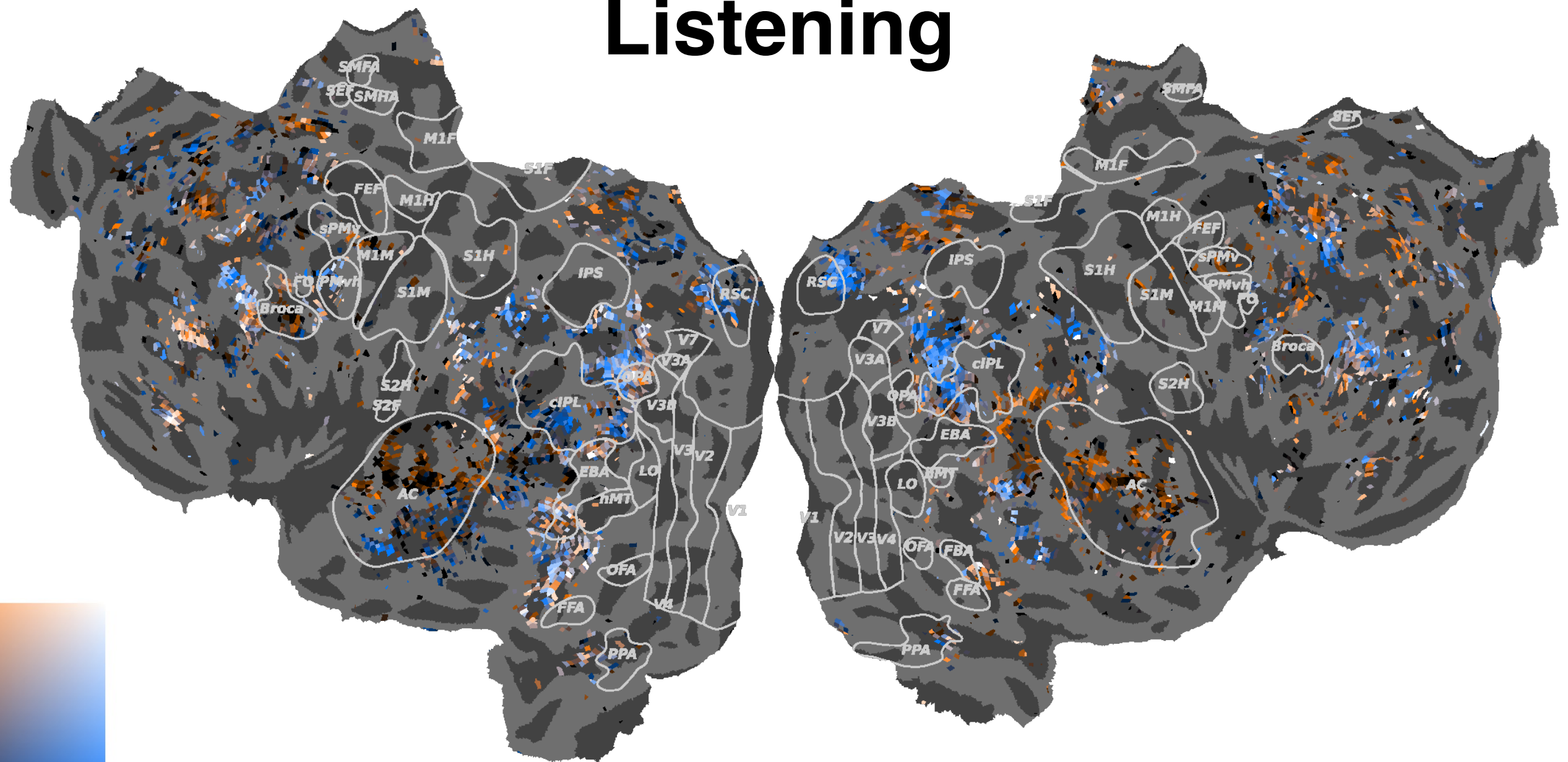
Subject 7
Baseline - Noun



Subject 7
Reading



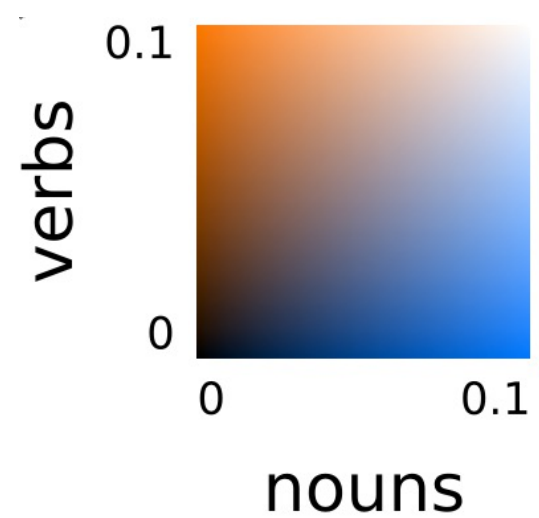
Subject 7
Listening



Technische
Universität
Berlin

Subject 7

Listening



COMPARISON WITH LITERATURE

Theoretical account	Predicted pattern	Observed pattern
Fronto-Temporal View <i>Neuropsychological</i> (A. R. Damasio and Tranel 1993)	V > N in frontal regions; N > V in temporal regions	V > N in temporal regions (ATL, PTL); no clean frontal/temporal split.
Emergentist View <i>Object-Action, Imageability</i> (Bird et al. 2000; Vigliocco et al. 2011)	V > N in frontal and parietal regions; N > V in inferior temporal regions.	N > V near visual areas; Similar sensitivity in frontal regions.
Combinatorial <i>Morphology/inflection</i> (Tyler et al. 2004; K. A. Shapiro et al. 2005)	N = V broadly, with V > N specifically in IFG for morphological composition.	Slight V > N in IFG; N > V in IFGOrb.
Distributed <i>Shared network</i> (Crepaldi et al. 2013; Caucheteux et al. 2021)	Noun and verb circuits distributed across frontal, parietal, and temporal regions.	Smaller clusters of noun and verb sensitive voxels across the brain.

LIMITATIONS

Only the *absence* of nouns / verbs is being investigated

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ROIs used in quantification don't cover all clusters /
Clusters split across multiple ROIs

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Cross-Modality *not confirmed*

Investigating *presence* of nouns and verbs

Using one-hot encodings

Investigating *presence* of nouns and verbs

Using one-hot encodings

Using a different dataset

Other than Moth

Investigating *presence* of nouns and verbs

Using one-hot encodings

Using a different dataset

Other than Moth

Looking at a different Language

Other than English, e.g. Chinese

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Cluster Categorization

Which kind of nouns / verbs in the clusters?